



PROBLEM SPACE

What problem are you solving?

USER Characteristics

What are the users like?

Group(s)		Needs
Who are the users? Are there supporting users? For example: students and teachers.		What do the users need?
primary users	ED triage nurses	- Fast, readable interface - Accurate AI recommendations - Minimal additional data entry - Clear explanation of recommendations - System must not slow triage
secondary users	ED physicians/consultants Clinical IT staff ED administrators	- Reliable triage information - Audit trail of AI recommendations and overrides - Secure handling of patient information - Consistent system performance
Goal(s)		
What do the primary and secondary users want to accomplish?		
primary users	- Assess patients quickly and accurately. - Identify high-risk patients early. - Receive clear AI decision support. - Maintain control of the final ESI decision	- Improve consistency of triage decisions. - Reduce missed high-risk cases. - Reduce cognitive workload during busy shifts. - Use AI safely within routine triage.
secondary users	- Receive clear, reliable triage information. - Identify urgent patients for faster review. - Understand the basis of AI-supported decisions.	- Improve patient flow and prioritisation. - Reduce preventable delays in urgent care. - Monitor system safety and performance. - Support accountable AI use in the ED.
SHORT-TERM		LONG-TERM

ROBOT

Task(s)

What task does the robot perform?

Task(s)			
What task does the robot perform?			
- Reliable triage information - Audit trail of AI recommendations and overrides - Secure handling of patient information - Consistent system performance		- Integrate with Mercer's EHR. - Retrieve available patient information automatically. - Maintain an audit trail. - Support ongoing model monitoring and evaluation.	
SHORT-TERM		LONG-TERM	
Advantages			
What advantages does using a robot bring (compared to a computer or human)?			
	Simple visual alerts help nurses recognise urgency quickly.	Recommendation based on the individual patient's clinical information.	Rapid, consistent processing of patient data.
Social skills	User's emotional response	Personalization	Precise tasks
			- Potential EHR integration. - Can store recommendations and nurse decisions.
Data collection with sensors	Mobility	Environment manipulation	Connection to systems



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ETHICAL CONSIDERATIONS

Consider potential ethical problems, and potential solutions –both from the user’s and robot’s perspectives.
Consider the boxes to be guidelines: you don’t need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

PROBLEM	USER		SOLUTION
	Nurse may rely on an incorrect AI recommendation.	Nurse reviews the recommendation and retains final decision authority.	
	ROBOT		
Incorrect predictions could contribute to delayed prioritisation.	AI remains advisory and provides clear Accept/Override controls.		

Data security

Is the robot in a unique data collection position? How is the user’s data protected?

PROBLEM	USER		SOLUTION
	Unauthorised staff could access sensitive patient information.	Restrict access to authorised clinical staff.	
	ROBOT		
Patient demographics and clinical data could be exposed or improperly stored.	Use secure authentication, encryption, minimal data collection and audit logs.		

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

PROBLEM	Nurse may not understand why the AI recommends a particular ESI level.	Show the recommendation, confidence and contributing clinical factors.	SOLUTION
	The AI prediction may appear more certain or authoritative than it actually is.	Clearly label predictions as clinical decision support, not final decisions.	

Equality across users

Robots’ algorithms can be biased. A robot’s appearance could reinforce harmful stereotypes. What are potential issues?

PROBLEM	Some patient groups may receive less reliable recommendations.	Nurses retain authority to override recommendations based on clinical judgement.	SOLUTION
	USER		
	ROBOT		
	Training data may underrepresent Caribbean or particular demographic groups.	Test performance across patient groups and conduct local validation before deployment.	

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

PROBLEM	USER		SOLUTION
	Patients may feel uncomfortable or believe AI is replacing their nurse.	Explain that AI only assists the clinical team.	
	ROBOT		
The interface could make the AI appear more authoritative than the clinician.	Use neutral clinical language and emphasise nurse control.		

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

PROBLEM	Repeated use could lead nurses to automatically accept AI recommendations.	Require active confirmation and provide an easy override option.	SOLUTION
	Interface design could encourage automation bias..	Avoid automatic acceptance and record overrides for monitoring.	
	USER		
	ROBOT		





DESIGN GUIDELINES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?
Think back to what you defined in the solution space canvas.

- Provide rapid and consistent triage support.
- Highlight high-risk patients clearly.
- Reduce unnecessary manual data entry where possible.
- Support, rather than replace, clinical judgement.
- Integrate with existing hospital systems.

Ethical guidelines

What ethical considerations does the robot have?
Think back to what you defined in the ethics canvas.

- Nurse must retain final decision authority.
- AI recommendations must be clearly identified as advisory.
- Protect patient confidentiality and clinical data.
- Monitor performance for demographic bias.
- Record AI recommendations, overrides and final decisions for accountability.

ROBOT DIMENSIONS

Environment guidelines

- Designed for a busy, interruption-heavy ED.
- Information must remain readable under time pressure.
- Require minimal steps to complete a triage assessment.
- Maintain a manual fallback during system/network failure.
- Avoid unnecessary alerts that contribute to alarm fatigue.

Form guidelines

What guides the design of the robot's outward qualities?
For example:

- Should the robot be designed to appear especially approachable, or more industrial?
- Should it be simple, or detailed?

- Clean, professional clinical appearance.
- Large, legible text and controls.
- Clear separation of patient data, vitals, AI output and nurse decision.
- Colour should support urgency but not be the only indicator.
- Avoid decorative elements that distract from clinical information.

Interaction guidelines

What guides the design of interaction?

For example:

- Is the interaction multimodal, or is one modality optimized for efficiency?
- Should the user feel empowered and lead the interaction, or does the robot provide safety via leadership?
- Is the goal of the interaction to complete a task, or explore?

- Nurse should lead the interaction.
- Accept and Override controls must be obvious.
- Require active confirmation before submitting a decision.
- Important information should be accessible at a glance.
- Minimise clicks and typing.

Behaviour guidelines

What guides the design of the robot's behaviour?

For example:

- Should behaviour be simple, or sensitive to context?
- Does the robot have internal drivers, or does it react to external stimuli?
- Does the robot have social skills?

- AI responds only to available clinical inputs.
- Do not automatically assign the final ESI level.
- Clearly flag abnormal or incomplete data.
- Show uncertainty rather than presenting predictions as certain.
- If the system fails, instruct the nurse to continue standard manual triage.





ROBOT DESIGN MVP

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

Where and when

Where: Mercer General Hospital ED triage desk
When: During patient triage, 24/7
Used after initial patient information and vital signs are available
Designed for both normal and high-volume periods

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

Robot's role

Clinical decision-support assistant, supports the triage nurse, Identifies potentially high-risk patient, Recommends an ESI level, Does not make the final triage decision

Personality

Its neutral and professional

Context-based behaviour

- Highlights abnormal vital signs.
- Adjusts recommendation using available patient data.
- Flags missing/incomplete inputs.
- Displays higher urgency clearly.
- Does not produce a final clinical decision without nurse confirmation.

Connection to systems

- EHR integration for patient information and available clinical data.
- Receives vital-sign data.
- Sends/stores AI recommendation and final nurse decision.
- Maintains an audit log of recommendations and overrides.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

- ☐ voice
☐ sounds
☐ gestures
☐ movement
☐ touch
☐ smell
☐ facial expression
☒ screens
☐ lights
☒ other
☒ Keyboard/mouse
☒ EHR data

OUTPUT

- ☐ voice
☐ sounds
☐ gestures
☐ movement
☐ touch
☐ smell
☐ facial expression
☒ screens
☒ lights
☒ other
☒ Visual Alerts

Interaction flow

Describe the most important interaction of the robot.

Note: only fill the bottom row if your robot is teleoperated.

ROBOT USER

OPERATOR
Robot
(optional)

BEFORE	DURING	AFTER
• Nurse begins patient assessment. • Reviews patient information. • Records/confirm vital signs and chief complaint.	• Reviews AI ESI recommendation. • Checks confidence and contributing factors. • Uses clinical judgement. • Accepts or overrides recommendation.	• Submits final triage decision. • Continues standard ED workflow. • Escalates urgent patient as required.
• Receives patient information and vital signs. • Checks required inputs.	• Processes patient data. • Predicts ESI level. • Displays confidence and contributing factors. • Highlights abnormal findings.	• Records recommendation and nurse decision. • Records override reason if applicable. • Updates/sends information to clinical system.



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ENVIRONMENT

What is the robot's context of operation?
You can use the "Ecosystem" canvas to dive deeper into this topic.

Where

What place?
Does it change?

- Mercer General Hospital ED triage desk
- Used within the existing triage workflow
- Busy, interruption-heavy clinical environment
- Fixed nurse workstation

User(s)

Who is using the robot?

- ED triage nurse
- Clinically trained
- Works under significant time pressure
- Responsible for the final triage decision

When

What time of day?
Does it change?

- During initial patient triage
- Available 24/7
- Used after patient information and vital signs are available
- Must function during both normal and high-volume periods

Secondary user(s)

Are there secondary users?
E.g. teachers that help students use a robot.

- ED physicians/consultants
- Nurse-in-charge
- Clinical IT staff
- ED administrators

Data collection

Does the robot collect data from its environment?
How is it stored?

- | | |
|------------------------|--|
| • Patient demographics | • Nurse's final decision |
| • Chief complaint | • Override reason, when applicable |
| • Vital signs | • Stored securely within authorised hospital systems |
| • AI recommendation | • Access controlled and auditable |



TRADE-OFF:

More data collection requires more attention to data security.

Simultaneous users

How many users should be able to use the robot simultaneously?



TRADE-OFF:

More simultaneous users requires a more sophisticated robot.



- One triage nurse interacts with a patient record at a time.
- Multiple authorised workstations could use the system independently.

External sensors and actuators

Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

- No robotic actuators required.
- May receive readings from existing clinical vital-sign devices.
- Keyboard, mouse or touchscreen used for interaction.
- System should not depend on specialised robotic hardware

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?

How does it use these systems?

- Electronic Health Record (EHR)
- Hospital patient database
- Existing vital-sign records/devices where integration is available
- Secure authentication system
- Stores AI recommendation and final nurse decision
- Maintains audit logs



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FORM

SOCIAL ROBOT CO-DESIGN CANVASES

What are the robot's outward qualities? If an existing robot is used, are its qualities modified?

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

MERCER GENERAL HOSPITAL
EMERGENCY DEPARTMENT

AI-ASSISTED ED TRIAGE DASHBOARD
Clinical Decision Support System

14:32
14 Mar 2023
Logged in as: Triage Nurse

PATIENT INFORMATION

Patient ID: #ED-20452
Name: J. Hartwell, M.
DOB / Age: 03/14/1971 (52)
Gender: Male
Arrival Time: 14:22
Chief Complaint: Chest pain, shortness of breath
Triage Status: **IN PROGRESS**
Pain Score: 8/10

VITAL SIGNS

Heart Rate (HR): **118** bpm
Blood Pressure (BP): **142/91** mmHg
Respiratory Rate (RR): **22** breaths/min
Temperature (Temp): **38.9** °C
SpO2 (Oxygen Saturation): **94** %
GCS (Glasgow Coma Scale): **14**

AI RECOMMENDATION

PREDICTED ESI LEVEL: **ESI 2**
CONFIDENCE SCORE: **87.4%**

CLINICAL DECISION

NURSE DECISION: **ACCEPT RECOMMENDATION** (Override Reason: Required if overriding)
Additional Notes (Optional):
SUMMIT DECISION

Clinical Decision Support Only - Final triage decision remains with the attending nurse. Patient data is secure and confidential.

Appearance

Is the robot more machine or lifelike? Is it human-shaped, animal-shaped, or abstract?

TRADE-OFF:

Robots that appear more human and lifelike are expected to be more sophisticated in features.

machine-like | humanoid | zoomorphic | lifelike

Size

- Professional clinical dashboard
- Simple, non-human appearance
- Clearly presented as an AI decision-support tool
- No human-like avatar or features

How big is the robot?

smaller | human-size | bigger

Character of movement

What is the robot's movement like?

No physical movement

machine-like | hybrid | lifelike

Voice & sounds

Does the voice have a gender or an age? What are pitch, speed and prosody like? Is the voice always the same?
Does the robot make sounds: music, "beep"s, animal noises?
When are these sounds heard?

- Primarily silent
- No voice interaction required
- Optional audio alert for critical cases only
- Alerts should be limited to prevent alarm fatigue

Mobility

Does the robot move across space? Does it move in place?

- Fixed at ED triage workstation
- Software could also run on an approved hospital tablet/workstation
- Does not physically move through the ED

Visual cues

Does the robot have expressions, lights, a screen or other visual elements?

- Clear ESI level display
- Colour-coded urgency indicators
- Abnormal vital signs highlighted
- Confidence score displayed clearly
- Large Accept and Override controls
- Colour is not the only way urgency is communicated

Touch & smell sensations

Is the robot soft or rough, warm or cold?

How does the robot smell?

Touch and smell are especially important in close interactions.

N/A — standard hospital computer/touchscreen; no specialised tactile or smell features.



Week 9 Final Deliverable:

HCI Co-Design Canvas

Problem Busy ED triage can lead to inconsistent ESI decisions. The AI provides decision support while nurses retain final authority.	Ethics Protect patient privacy, reduce bias, ensure transparency, maintain clinician oversight.	Guidelines Simple interface, readable under pressure, minimal clicks, clear colour coding.
MVP Patient form, Predict button, ESI prediction, confidence score, override button.	Environment Busy emergency department, noisy, interruptions, multiple clinicians.	Form Desktop/web dashboard used at the ED triage desk.

HRI Co-Design Canvas

Problem Observation Unit patients require repeated monitoring and early deterioration detection.	Ethics Patient safety, privacy, human supervision, reliable sensor behaviour.	Guidelines Safe interaction, manual fallback, simple prompts, clear alerts.
MVP Robot kiosk with vitals collection and AI recommendation.	Environment Observation Unit with moderate noise and continuous monitoring.	Form Robot-assisted kiosk with touchscreen and vital-sign devices.

Preferred Implementation: HCI

Problem: The Emergency Department requires a fast, consistent, and transparent triage process that supports clinicians during periods of high patient demand.

Ethics

- Protect patient confidentiality.
- Reduce algorithmic bias.
- Maintain transparency.
- Ensure clinicians remain responsible for final decisions.
- Record overrides for accountability.

Guidelines

- Simple interface.
- Clear visual hierarchy.
- Minimal clicks.
- Readable under stressful conditions.
- Fast response time.

Minimum Viable Product (MVP)

The initial system includes:

- Patient information form.
- AI prediction button.
- ESI recommendation.
- Confidence score.
- Override option.
- Audit log.
- Environment

The system will operate within:

- Emergency Department triage desk.
- High patient volumes.
- Frequent interruptions.
- Busy and noisy clinical environment.

Form: The preferred implementation is a desktop/web dashboard used by triage nurses during patient assessment.

Guidelines

- Simple interface.
- Clear visual hierarchy.
- Minimal clicks.
- Readable under stressful conditions.
- Fast response time.

Minimum Viable Product (MVP)

The initial system includes:

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The system will operate within:

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- High patient volumes.
- Frequent interruptions.
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Form: The preferred implementation is a desktop/web dashboard used by triage nurses during patient assessment.

Mercer General Hospital

AI Triage Assistant

AI-assisted decision support for ED triage

Triage Nurse

15 May 2025 10:24 AM

1. PATIENT INFORMATION

Demographics

Patient ID

MGH-2025-0515-1024

Age

63

years

Sex

Female

Presenting Complaint

Shortness of breath and chest tightness for 2 hours.

Vital Signs

Heart Rate

118

bpm

Respiratory Rate

24

breaths/min

Blood Pressure

144 / 92

mmHg

Temperature

37.8

°C

Oxygen Saturation (SpO₂)

93

%

Glasgow Coma Scale (GCS)

15

/15

Clear

Save Patient

2. AI TRIAGE RECOMMENDATION

Predicted ESI Level

ESI 2

High Acuity – Emergent

Confidence Score

0.87 (87%)

Explanation

The patient's vital signs indicate possible cardiovascular or respiratory compromise.

- Elevated heart rate (118 bpm)
- Elevated respiratory rate (24 breaths/min)
- High blood pressure (144/92 mmHg)
- Low-normal oxygen saturation (93%)

These factors are associated with higher urgency.

3. CLINICAL ACTION

AI Recommendation

ESI Level: 2 – High Acuity

Recommended Location: Resuscitation / Majors

Your Decision

Accept Recommendation

Override Recommendation

Override Reason (if applicable)

Select reason...

Additional notes...

Submit Decision

Recent History

No recent predictions for this patient.

All patient data is secure and confidential.

Audit Log

Logout

MERCER GENERAL HOSPITAL
EMERGENCY DEPARTMENT

AI-ASSISTED ED TRIAGE DASHBOARD
Clinical Decision Support System

14:32
14 Mar 2025

Logged in as:
Triage Nurse

PATIENT INFORMATION

Patient ID: #ED-20482
 Name: J. Hartwell, M
 DOB / Age: 03/14/1971 (53)
 Gender: Male
 Arrival Time: 14:32
 Chief Complaint: Chest pain, shortness of breath
 Triage Status: ● IN PROGRESS
 Pain Score: 8/10

VITAL SIGNS

Heart Rate (HR) 118 bpm ↑ HIGH	Blood Pressure (BP) 142/91 mmHg ↑ HIGH	Respiratory Rate (RR) 22 breaths/min ↑ HIGH
Temperature (Temp) 38.9 °C ↑ HIGH	SpO ₂ (Oxygen Saturation) 94 % ↓ LOW	GCS (Glasgow Coma Scale) 14 ✓ NORMAL

AI RECOMMENDATION

PREDICTED ESI LEVEL: **ESI 2**
EMERGENT

CONFIDENCE SCORE: **87.4%**

TOP FACTORS (Contribution to Prediction):

- Tachycardia (HR 118 bpm)
- Elevated Blood Pressure (142/91 mmHg)
- Fever / High Temperature (38.9 °C)
- Low Oxygen Saturation (94%)
- Chief Complaint: Chest pain, shortness of breath

Model: TriageAI v2.3 | Last Updated: 14:33:05

CLINICAL DECISION

NURSE DECISION:

☒ ACCEPT RECOMMENDATION
 ☐ OVERRIDE RECOMMENDATION

Override Reason (Required if overriding):

Select reason

Additional Notes (Optional):

Enter notes here...

SUBMIT DECISION

Clinical Decision Support Only — Final triage decision remains with the attending nurse.
 Patient data is secure and confidential.

Primary User

Emergency Department Triage Nurse

The triage nurse is the primary user of the system. The nurse will review patient information and vital signs, receive the AI-generated ESI recommendation, and make the final clinical triage decision.

Secondary Users:

- Emergency Department Physicians/Consultants
- Nurse-in-Charge
- Clinical Managers
- Hospital IT Staff

Secondary users may use information produced by the system for patient management, clinical oversight, system monitoring, governance, and technical support.

2. Selected Implementation

The selected implementation is a Human–Computer Interaction (HCI) web-based dashboard located at the Mercer General Hospital Emergency Department triage desk.

The dashboard will provide AI-assisted Emergency Severity Index (ESI) recommendations to triage nurses while maintaining human oversight. The system is intended to support clinical decision-making rather than replace it. The triage nurse remains responsible for the final triage decision.

A Human–Robot Interaction (HRI) implementation was also considered for the Observation Unit. This would involve a robotic kiosk or vitals station capable of collecting patient vital signs and providing information to the AI system. However, the HCI dashboard was selected as the preferred implementation because it can integrate more easily into the existing ED triage workflow and requires less additional physical infrastructure.

3. System Inputs

The system receives information collected during patient triage.

Manual Entry

The initial implementation will support the entry of:

1. Age
2. Sex
3. Presenting complaint
4. Heart rate

5. Respiratory rate
6. Blood pressure
7. Temperature
8. Oxygen saturation (SpO₂)
9. Glasgow Coma Scale (GCS)

These inputs represent patient information and physiological measurements available during the triage process.

Future Inputs

Future versions of the system could receive information from:

1. Electronic Health Record (EHR)
2. Vital-sign monitoring devices
3. Robotic sensor station in the Observation Unit

These future input methods would reduce manual data entry and allow information to be transferred directly from existing clinical systems or medical devices.

4. System Processing

After receiving the required patient information, the system will:

- Validate the entered patient information and vital signs.
- Identify missing or invalid required information.
- Process the clinical information required by the AI model.
- Generate a predicted ESI category.
- Generate a confidence score for the recommendation.
- Present the prediction and relevant information to the triage nurse.
- Allow the nurse to review, accept, or override the recommendation.
- Record the final clinical decision.

The AI model will therefore operate as a clinical decision-support system, rather than as an autonomous triage system.

5. System Outputs

The system will produce:

- Predicted ESI level (1–5)
- Confidence score
- Colour-coded urgency indicator
- Key factors influencing the prediction
- Relevant abnormal vital-sign indicators
- Recommendation for clinician review
- Warning when required information is missing or invalid

The nurse will then have the option to:

- Accept the recommendation
- Override the recommendation
- Record a reason for overriding
- Submit the final triage decision

The AI-generated ESI level represents a recommendation only. It does not automatically become the patient's final triage classification.

6. Human Action Required

After receiving the AI recommendation, the triage nurse will:

- Review the predicted ESI level.
- Review the confidence score.
- Review the patient's vital signs and relevant contributing factors.
- Compare the AI recommendation with their own clinical assessment.
- Accept the recommendation if clinically appropriate or override it if necessary.
- Record a reason for an override where applicable.
- Submit the final triage decision.
- Continue the normal ED workflow and escalate the patient where clinically required.

The nurse therefore remains in the loop throughout the decision-making process and retains final clinical authority.

The system will produce:

- Predicted ESI level (1–5)
- Confidence score
- Colour-coded urgency indicator
- Key factors influencing the prediction
- Relevant abnormal vital-sign indicators
- Recommendation for clinician review
- Warning when required information is missing or invalid

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- Review the confidence score.
- Review the patient's vital signs and relevant contributing factors.
- Compare the AI recommendation with their own clinical assessment.
- Accept the recommendation if clinically appropriate or override it if necessary.
- Record a reason for an override where applicable.
- Submit the final triage decision.
- Continue the normal ED workflow and escalate the patient where clinically required.

The nurse therefore remains in the loop throughout the decision-making process and retains final clinical authority.

7. Functional Requirements

The system shall:

- Accept patient information.
- Accept required clinical and vital-sign information.
- Validate entered data.
- Identify missing or invalid required information.
- Predict an ESI category.
- Generate a confidence score.
- Display prediction results clearly.
- Display relevant factors influencing the recommendation.
- Highlight important abnormal clinical information.
- Allow nurses to accept recommendations.
- Allow nurses to override recommendations.
- Allow an override reason to be recorded.
- Record overrides for audit purposes.
- Save prediction history.
- Record the final nurse decision.
- Handle missing data gracefully.

8. Non-Functional Requirements

Performance

- The system should produce predictions in under five seconds so that it does not create a significant delay during triage.

Usability

- The system should provide a simple and intuitive interface suitable for use in a busy Emergency Department.
- Important clinical information should be visible without excessive navigation or unnecessary clicks.

Availability

- The system should be available throughout clinical shifts.

- Failure of the AI system must not prevent clinicians from continuing standard manual triage.

Security and Privacy

- The system must maintain patient confidentiality.
- Access to patient information should be restricted to authorised personnel, and clinical information should be securely transmitted and stored.

Accessibility

The interface should use:

- Clear labels
- Readable text
- Appropriate contrast
- Large and obvious controls
- Multiple methods of communicating urgency

Explainability

- The system should provide relevant information about its recommendation rather than displaying only an unexplained ESI category.
- This may include the confidence score and important clinical factors associated with the recommendation.

Auditability

- AI recommendations, nurse overrides, override reasons and final clinical decisions should be recorded where appropriate to support monitoring and clinical governance.

Scalability and Integration

- The system should support future integration with hospital information systems and clinical devices.

9. Integration Requirements

The system should support future integration with:

- Hospital Electronic Health Record (EHR)
- Patient registration system
- Vital-sign monitoring devices
- Hospital authentication system

Integration would allow patient information and clinical measurements to be transferred to the system with less manual entry.

During an initial pilot, manual entry may still be required while integration capabilities are evaluated.

10. HCI Safety Considerations

Automation Bias

- Risk: Nurses may begin to rely excessively on AI recommendations rather than independently assessing the patient.
- Mitigation: The system should require active nurse confirmation and provide a clear override option. The interface should state that the recommendation is clinical decision support only.

Alarm Fatigue

- Risk: Excessive warnings and alerts could cause clinicians to ignore important notifications.
- Mitigation: Alerts should be limited to clinically meaningful situations and prioritised according to urgency.

Interface Legibility Under Stress

- Risk: Small text, excessive information or unclear controls may increase the chance of user error during busy periods.
- Mitigation: The dashboard should use a clear visual hierarchy, readable text, simple controls and prominent urgency indicators.

11. HRI Safety Considerations

Although the HCI dashboard is the preferred implementation, an HRI implementation was considered for the Observation Unit.

Sensor Failure

- Risk: A robotic vitals station could produce inaccurate or missing measurements.
- Mitigation: The system should detect implausible or missing readings, alert clinical staff and allow vital signs to be measured manually.

Physical and Proximity Safety

- Risk: Patients and staff may physically interact with robotic equipment.
- Mitigation: Robotic equipment should use safe positioning and controlled movement, and appropriate emergency-stop or manual shutdown mechanisms should be available where necessary.

Noisy Clinical Environment

- Risk: Voice or audio-based interaction may be unreliable in a busy clinical environment.
- Mitigation: The robotic system should not depend exclusively on voice interaction. Visual and touchscreen alternatives should be available, with clinical staff able to assist when necessary.

12. Failure and Fallback Requirements

If the AI model, interface, network connection or connected hospital system becomes unavailable:

- The nurse should be clearly informed that the AI recommendation is unavailable.
- The system should not display an unsupported or incomplete recommendation.
- Standard manual triage procedures should remain available.
- System failure must not prevent urgent clinical assessment.
- Missing information should be clearly identified rather than silently estimated where this could compromise safety.
- Technical failures should be recorded where appropriate for investigation.

13. Limitations

Current limitations include:

- The model uses retrospective training data.
- The AI system does not replace clinical judgement.
- Manual data entry may be required during the pilot phase.
- The model requires further validation before deployment.
- Performance may not generalise equally across all patient populations.
- Full EHR and medical-device integration has not yet been implemented.
- The proposed interface is currently a prototype rather than a deployed clinical system.

14. Future Improvements

Future work may include:

- Live EHR integration
- Automated vital-sign collection
- Observation Unit robotic kiosk
- Continuous model monitoring
- Explainable AI dashboard
- Automated retrieval of available patient information
- Performance monitoring across patient demographic groups
- User testing with ED nurses
- Evaluation of the system within the actual Mercer triage workflow

15. Summary

The proposed system is a nurse-facing HCI clinical decision-support dashboard designed to assist with emergency department triage at Mercer General Hospital.

The system receives patient information and vital signs, generates an AI-assisted ESI recommendation and presents the result to the triage nurse. The nurse reviews the recommendation alongside the patient's clinical information and retains responsibility for the final triage decision.

Video Link:

<https://www.loom.com/share/fa05591a95544885b0854865038b32d0>